What Missionary Control Left in the Language of Two 1880s isiXhosa Newspapers

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Abstract

From 1884 the missionary *Isigidimi sama-Xosa* and the African-owned *Imvo Zabantsundu* ran side by side. Did missionary control change *what* they printed, or *how* they used words? Imvo was not more political: of ten political topics chosen in advance, only parliament differs, and not after correction. Religion, an exploratory topic, differs most: Isigidimi uses it almost four times as often, also in readers' letters and under the same editor (also uncorrected). How political words were used barely differs. Old spelling costs only the African-language tokenizer more. The evidence fits control shaping what the papers printed; any effect on how they wrote is below what we can detect.

Introduction and Background

Where did missionary control leave its trace in the language of the earliest isiXhosa press? The IsiXhosa Intellectual Traditions archive (Schoots et al., 2023a) lets us ask this. *Isigidimi sama-Xosa* was a missionary paper. *Imvo Zabantsundu* was founded in 1884 by Isigidimi's former editor, J. T. Jabavu, as the first African-owned isiXhosa paper, free "from any missionary control" (Schoots et al., 2023a, p. 55). After 1884 both papers had African editors, Gqoba and Jabavu (Schoots, 2023b, p. 71). But comparing the papers also compares two editors, places and readerships, so we also compare Jabavu with himself: he edited Isigidimi from 1881 to 1884 and Imvo afterwards.

Communication research distinguishes what a paper covers, *agenda-setting* (McCombs & Shaw, 1972), from how it presents it, *framing* (Entman, 1993). We measure coverage as **salience**: how often a topic's words appear. We study framing with word embeddings, which place each word by its neighbours (Hamilton et al., 2016a). Most such studies use modern news (Card et al., 2015; Field et al., 2018); Schoots (2023b) counted root words in these papers. We add controlled comparisons, a same-editor test and a letters-versus-rest split, and ask whether this history affects how machines read the archive (Petrov et al., 2023; Rust et al., 2021). **H1**: the papers differ in how often they use political and religious words. **H2**: Isigidimi uses political words in more religious contexts. **H3**: old text and spelling cost tokenizers more.

Methods

Data. We use the archive's human-corrected transcriptions, which were not spell-checked (Schoots et al., 2023a, p. 59), from November 1884 to 1888: 286,663 words of Isigidimi and 1.31 million of Imvo (safeguards in Table 1).

Salience. Each topic is a set of word stems, adapted from Schoots (2023b), including spelling variants such as Imvo's *rulumeni* for *rulumente*, 'government' (Table S1). We fixed ten political topics in advance, plus four neutral controls. Religion (*tixo*, *kristu*) is exploratory: a classifier trained on the same years suggested it (logistic regression, issue-grouped nested cross-validation, 0.83 balanced accuracy; chance 0.50), though the archive's authors also expect it (Schoots et al., 2023a, p. 57). For rates per million words, bootstrap confidence intervals over whole issues test whether the rates differ; matched random word sets ask the stricter question: does it differ more than random words of the same frequency? For the same-editor contrast, Jabavu's Isigidimi runs January 1881–October 1884.

Threat	Response
Different years, sizes	Nov 1884–88; ten month- and size-matched draws
Frequency effects	2000 matched random word sets (Monroe et al., 2008)
Change over time	Each paper's own drift in other years
Unstable embeddings	Tune for seed stability; average ten runs
Non-political effects	Four neutral control concepts
One paper per side	Same editor in both; place, audience unfixed

Table 1: Threats to a fair comparison, and our response.

Letters. A reader's letter opens by addressing the editor (*nkosi mhleli . . . kaundifakele*, 'sir, editor . . . please print'). In a hand check, 28 of 30 detected letters were genuine (Table S5).

Framing. Embeddings capture a word's meaning, grammar and topic, so they are only a rough proxy for framing in Entman's sense. We train fastText embeddings (Bojanowski et al., 2017), tuning 34 settings (Table S8) on other years for stability across random seeds, since unstable embeddings create false shifts (Antoniak & Mimno, 2018). This narrow window favours grammar, so we also report wider ones. A word's *shift* measures how differently it relates to 500 common words in each paper (Hamilton et al., 2016b); shuffling issues between the papers gives the level expected by chance (Dubossarsky et al., 2017). A directional test, the part aimed at framing, asks whether political words sit closer to religious or political words (Table S4). We also count religious and legal words near *umhlaba*.

Tokenizers. We count the subword pieces that XLM-R (Conneau et al., 2020), Serengeti (Adebara et al., 2023) and mBERT (Devlin et al., 2019) need for modern news (Adelani et al., 2023), for the same news in old spelling, and for our historical text.

Results

H1: different topics, but not more politics. Of the ten political topics, only parliament beats random words: Imvo uses it 3.3 times as often ($p = 0.016$; Fig. 1, left). Imvo uses law, rule and government more too, but not beyond random words; vote, rights and land do not differ. Religion differs most:

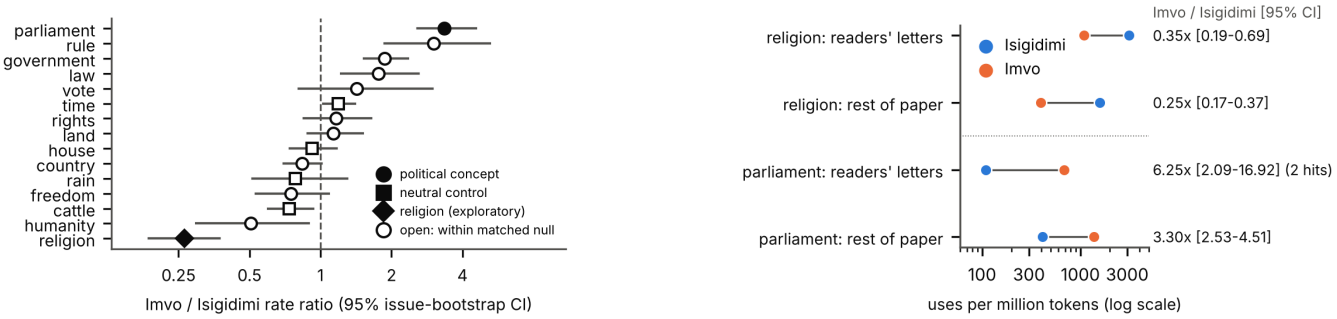


Figure 1: What the papers printed. Left: lmvo's rate relative to Isigidimi's, per topic (95% CI; filled: beats random word sets, $p < 0.05$ uncorrected). Right: the religious gap also appears in letters.

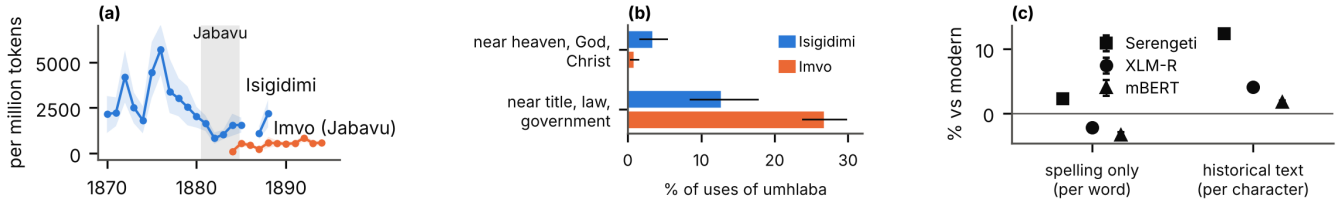


Figure 2: (a) Religious words per million; grey: Jabavu edits Isigidimi, then lmvo; Gqoba edits Isigidimi from late 1884. (b) *Umhlaba* near religious or legal words. (c) Extra subword pieces against modern text (CIs smaller than markers).

Isigidimi uses it 3.8 times as often ($p = 0.019$). Neither result survives correction for 15 tests. The religious gap appears in letters as well as the rest of each paper (Fig. 1, right). Under one editor, Jabavu's lmvo used religious words a third as often as his Isigidimi ($p = 0.025$). Under Gqoba, Isigidimi's rate was 1.38 times Jabavu's (CI 1.03–1.83, $p = 0.168$): no further fall, though the interval is wide. Before lmvo existed, Isigidimi's religious words were already falling, while parliament rose 50-fold and vote 12-fold ($p \leq 0.004$; Fig. 2a; Table S7).

H2: little difference in how words are used. Overall word use differs (mean shift 0.348, against 0.230 when issues are shuffled), but no topic stands out: parliament shifts no more than the neutral control *cattle* (Fig. S1). Political words do sit closer to religious words in Isigidimi in all five settings (windows 1, 2 and 5, with and without joined word forms), and in three after correction (Table S2, with neighbour overlap; Gonen et al., 2020). But this depends on one word, *land* (*umhlaba*), and the neutral control *house* shows the same lean; dropping the anchor *endlu* changes neither. The papers also use *umhlaba* differently: Isigidimi puts it near religious words in 3.4% of uses, lmvo in 0.9% (Fig. 2b). A single word must shift by 0.15 to be detected, more than the whole gap between the papers (0.118).

H3: historical text costs more. Our historical text needs 12.4%, 4.1% and 1.7% more subword pieces per character with Serengeti, XLM-R and mBERT; old spelling alone (per word) raises only Serengeti's cost (Fig. 2c).

Discussion

Is it control, or just time? (H1). Religious words were falling in Isigidimi anyway, so we combine two comparisons with different blind spots. lmvo against Gqoba's Isigidimi holds the years fixed; Jabavu's two papers hold the editor fixed. Both show the gap, and the second uses years the classifier never saw, partly replicating the religion finding. If the gap were only the falling trend, Isigidimi would have kept falling after 1884; we find no further fall. Had Jabavu's lmvo been as religious as his Isigidimi, control would be refuted; it was not. The argument rests on these two comparisons agreeing, not on their uncorrected p-values. For Jabavu, leaving brought less religion, not clearly more politics.

The gap is not only the editors' voice, or the sections each paper ran. Isigidimi's letter-writers used religious words three times as often as lmvo's, so each paper drew on, or chose, different writers. One wrote to Isigidimi: *ilizwi alizanga ukuba silipate, lize ukuba lisipate* ('the Word did not come for us to carry it; it came to carry [or rule] us'; 1884; our gloss).

What the embeddings show (H2). The framing signal looks like a difference in meaning, not slant: Isigidimi more often uses *umhlaba* for the earth, lmvo for land under law, which alone would move the word in each model. The detection floor rules out large framing differences, not small ones.

Responsible NLP (H3). History still affects today's tools: written isiXhosa took shape in the mission press (cf. Gilmour, 2007), and its old spelling penalises only the model built for African languages. Serengeti's vocabulary has twice XLM-R's share of *th*, *kh* and *ph* pieces (1.9% vs 0.9%), a possible reason. A modern-spelling copy can help tools but must not replace the original: conversion loses information (old *t* can be *t* or *th*). We describe two papers, not isiXhosa speakers; cultural claims without speakers would be extractive (Bird, 2020).

Limitations and conclusion. Two papers, one period, a missing year (1886), no African-owned paper under Gqoba, an automatic letter split, unreviewed translations. Still, the evidence fits control shaping what the papers printed; any effect on how they wrote is below what we can detect.

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Contribution statement

All three authors contributed equally to every stage of the project. Each stage was divided into three parts, one per author, as listed below; the decisions within each stage were made jointly, and every author reviewed the others' parts.

Stage	**Emmanuel Azubuike** (2698752)	**George Ayensu** (2722188)	**Banele Mkhize** (2663814)
Problem formulation and background	Research question and hypotheses H1–H3	Literature on the archive and these newspapers	Threats to a fair comparison and the response to each (Table 1)
Data preparation and exploration	Loading and cleaning both newspapers	Date- and size-matched sampling; concept word stems	Exploratory classifier and yearly series
Salience analysis (H1)	Rate ratios, bootstrap CIs and matched-set null	Readers' letter detection and letters-versus-rest test	Era drift and the same-editor contrasts
Framing analysis (H2)	Hyper-parameter search for stable embeddings	Profile-shift and neighbour-overlap measures; permutation null	Directional anchor test, robustness checks, detection floor and *umhlaba* senses
Tokenizer fairness (H3)	Tokenizer cost on modern and spelling-altered text	Tokenizer cost on historical text	Tokenizer vocabulary analysis
Responsible NLP and ethics	Limits of interpretation and speaker review	Spelling normalisation and archive labelling	NeurIPS checklist and AI declaration
Writing and reproducibility	Introduction, background and methods	Results, figures and tables	Discussion, supplement, code documentation and reproducibility checks

All authors reviewed and approved the final report and code, and take joint responsibility for them.

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Supplementary material

S1. Concepts, stems and the word forms they match (1 Nov 1884–1888, both papers)

Stems are matched as substrings of surface forms; following Schoots (2023b) we inspected forms for false positives (our first *law* stem also matched *inteto*, speech, and was dropped) but did not hand-filter them, so run-together forms (e.g. *rulumenteningomsebenzi*) and missed variants remain. All stems, forms and glosses await review by a first-language isiXhosa speaker; e.g. *rulumente* is glossed 'government' here but 'parliament' by Schoots et al. (2023a, p. 62). Political concepts were fixed before testing; religion was suggested by the classifier and is exploratory.

Concept	Type	Stems	Most frequent matched forms (count; number of distinct forms)
parliament	political	palamente	palamente (1754), ipalamente (47), palamenteni (22), epalamente (14), bapalamente (1), nepalamente (1), yepalamente (1); 11 forms
vote	political	voti	ivoti (301), voti (217), evotini (136), abavoti (98), ngevoti (55), levoti (22), nevoti (22); 115 forms
government	political	rulumente, rulumeni	rulumente (1882), rulumeni (1336), rulumenteni (7), urulumente (5), urulumeni (3), rulumenini (2), serulumente (1); 14 forms
rule	political	mbuso	zombuso (169), umbuso (76), mbuso (60), yombuso (42), kombuso (37), lombuso (28), kulombuso (23); 43 forms
law	political	mteto	umteto (724), lomteto (475), mteto (406), ngomteto (167), ngumteto (94), komteto (76), nomteto (76); 82 forms
rights	political	lungelo	amalungelo (315), ilungelo (288), lungelo (113), nelungelo (54), namalungelo (50), lilungelo (39), elilungelo (37); 115 forms
country	political	lizwe	lizwe (646), ilizwe (292), kwelilizwe (201), elilizwe (101), elizwe (96), belilizwe (76), yelilizwe (52); 78 forms
land	political	mhlaba	umhlaba (836), mhlaba (208), lomhlaba (156), nomhlaba (105), komhlaba (99), somhlaba (95), yomhlaba (94); 135 forms
freedom	political	kululek	inkululeko (27), kukululekile (24), ukululekile (13), ngenkululeko (10), abakululekileyo (8), nkululeko (8), nenkululeko (7); 92 forms
humanity	political	buntu	ubuntu (45), yobuntu (14), nobuntu (8), sobuntu (8), buntu (7), lobuntu (7), zobuntu (6); 29 forms
time	neutral	xesha	ixesha (787), ngexesha (364), xesha (298), amaxesha (243), lixesha (191), ngamaxesha (149), zamaxesha (114); 207 forms
house	neutral	ndlu	indlu (719), izindlu (328), kwindlu (276), ndlu (234), yendlu (227), ngendlu (133), nendlu (130); 193 forms
rain	neutral	mvula	imvula (267), mvula (31), yemvula (23), ezimvula (17), lemvula (16), yimvula (14), zimvula (10); 45 forms
cattle	neutral	nkomo	inkomo (555), zenkomo (258), nkomo (193), yenkomo (93), nenkomo (75), ngenkomo (35), lenkomo (33); 79 forms
religion	exploratory	tixo, kristu, krestu	tixo (684), kristu (207), krestu (118), makristu (7), kristukazi (7), makrestu (6), mkrestu (5); 32 forms

S2. Secondary framing tests across settings

Percentile of the observed shift among 2000 frequency-matched word sets, and one-sided p (uncorrected). Jaccard = shift in a word's ten nearest neighbours (neighbour overlap; Gonen et al., 2020). P+V = *palamente* + *voti*; P+V+R adds *rulumente* (uncertain meaning). Directional Δ: Isigidimi minus Imvo, religious minus political similarity (in z-score units), political concepts, two-sided p against matched sets; the last columns give the control *house*. Three settings survive Bonferroni over five ($p < 0.01$): window 1, window 2, and window 2 with forms joined. Joining partly disjunctive forms (*uku ba*) leaves the global profile gap unchanged. Matched sets are re-drawn for each test; without *land* the pooled set is smaller, so lower power may contribute to the higher p.

Setting	P+V profile	P+V Jaccard	P+V+R profile	P+V+R Jaccard	Δ	Δ p	p w/o land	house Δ	house p
window 1 (tuned)	92 (p=0.083)	96 (p=0.039)	92 (p=0.077)	99 (p=0.007)	+0.68	<0.001	0.010	+1.08	<0.001
window 1, forms joined	99 (p=0.010)	95 (p=0.053)	100 (p=0.001)	100 (p=0.002)	+0.44	0.016	0.088	+1.15	<0.001
window 2	86 (p=0.139)	94 (p=0.061)	90 (p=0.096)	99 (p=0.010)	+0.76	<0.001	0.045	+1.34	<0.001
window 2, forms joined	91 (p=0.094)	98 (p=0.023)	99 (p=0.011)	100 (p=0.002)	+0.58	0.009	0.064	+1.51	<0.001
window 5	62 (p=0.382)	98 (p=0.016)	81 (p=0.189)	100 (p=0.002)	+0.66	0.015	0.256	+0.84	0.061

S3. Era drift: does a concept's rate move across eras within one paper?

Per-million-token rates and ratios (later/earlier, or Imvo/Isigidimi) with issue-bootstrap 95% CIs.

Concept	Comparison	First	Second	Ratio (95% CI)
parliament	Isigidimi: later / earlier era	214	394	1.84 (0.96–4.98)
parliament	Imvo: later / earlier era	1320	758	0.57 (0.50–0.66)
parliament	Imvo 1889-94 / Isigidimi 1876-83 (held-out)	214	758	3.54 (1.98–9.58)
government	Isigidimi: later / earlier era	2197	1095	0.50 (0.39–0.64)
government	Imvo: later / earlier era	2234	1649	0.74 (0.67–0.81)
government	Imvo 1889-94 / Isigidimi 1876-83 (held-out)	2197	1649	0.75 (0.65–0.88)
vote	Isigidimi: later / earlier era	72	614	8.55 (3.51–18.94)
vote	Imvo: later / earlier era	799	921	1.15 (0.88–1.51)
vote	Imvo 1889-94 / Isigidimi 1876-83 (held-out)	72	921	12.82 (7.85–25.92)
religion	Isigidimi: later / earlier era	2541	1678	0.66 (0.50–0.86)
religion	Imvo: later / earlier era	445	612	1.38 (1.08–1.76)
religion	Imvo 1889-94 / Isigidimi 1876-83 (held-out)	2541	612	0.24 (0.20–0.30)

S4. Anchors for the directional test

Anchors are the 15 top-PMI neighbours of each seed pair in held-out text (Isigidimi 1876-83 + Imvo 1889-94), frozen before testing. Religious (15 words; seeds *tixo*, *kristu*): *sikelela*, *sindisa*, *ukumkanikazi*, *inkosazana*, *buka*, *oyingcwele*, *lilizwi*, *msindisi*, *nelizwi*, *kwelizwi*, *wanga*, *lika*, *yesu*, *izono*, *yehova*. Political (15 words; seeds *palamente*, *voti*): *emqulwini*, *wonyulo*, *lonyulo*, *elugqatsweni*, *eziqingata*, *lokunyula*, *kwesase*, *lokuvota*, *endlu*, *kwamagama*, *ukumela*, *izayo*, *ngalomteto*, *nawupina*, *kwabase*. Seed words cannot themselves be tested (Fig. S1). Political anchors include *endlu* ('in the house'), which might have favoured the neutral control *house*; re-running without *endlu* leaves both the political lean and *house*'s lean in place (outputs/framing_matched_tests_noendlu.csv), and *ngalomteto*, which contains the *law* stem. Glosses are unverified: two religious anchors (*ukumkanikazi*, *inkosazana*) may reflect royal rather than religious register, and several anchors on both sides (*lika*, *buka*; *nawupina*, *kwabase*, *izayo*) may be grammatical rather than topical, consistent with narrow-window PMI capturing grammar. All await speaker review.

S5. Readers' letters versus the rest of each paper

Letters are detected by src/12_letters.py: a letter opens by addressing the editor (vocative *mhleli* or *mcokeli*, not *u/ku/ngu mhleli*, which refer to him) or by asking for insertion (*ndifakele*, *kaundifakele*), and ends at the signature (a run of very short lines). A hand check of 30 random detections found 28 genuine letters; the errors were usifake umkala (horse advert: 'put a bit on it'); no mhleli wepepa lama bhulu ('and the editor of the Boer paper'). Recall was not measured. Imvo is restricted to the months in which Isigidimi has text; CIs from 4000 issue resamples; rates are not tested against matched sets. Hits = concept uses.

Part	Concept	Isig. tokens	Isig. hits	Imvo tokens	Imvo hits	Imvo/Isig.	95% CI
letters	parliament	18,344	2	70,475	48	6.25	2.09–16.92
letters	religion	18,344	57	70,475	77	0.35	0.19–0.69
letters	vote	18,344	4	70,475	58	3.77	1.04–18.93
letters	government	18,344	13	70,475	161	3.22	1.67–9.62
rest of paper	parliament	268,319	111	936,998	1280	3.30	2.53–4.51
rest of paper	religion	268,319	424	936,998	370	0.25	0.17–0.37
rest of paper	vote	268,319	172	936,998	821	1.37	0.79–2.83
rest of paper	government	268,319	301	936,998	1906	1.81	1.47–2.28

S6. Which sense of *umhlaba*?

For every use of the land stem (*mhlaba*) we look at the 10 tokens either side. Religious cues: *izulu*, *lizulu*, *nezulu*, and words containing *zulwini*, *tixo*, *kristu*, *yesu*, *msindisi*, *ngcwele* (bare *zulu* is excluded: *kwa zulu* is Zululand). Property/political cues: words containing *taitile*, *mteto*, *rulumente*, *palamente*, *mantyi*, *fama*, *voti*, *nyulo*, *rafu*, *lokishi*. Most uses have neither cue. Example lines for a speaker to check are in outputs/land_kwic.md.

Paper	Cue	Uses	Share	95% CI
Imvo	religious	1408	0.9%	0.4–1.4%
Imvo	property/political	1408	26.8%	23.8–29.7%
Isigidimi	religious	353	3.4%	1.7–5.3%
Isigidimi	property/political	353	12.7%	8.5–17.7%

S7. Same editor, different control; same control, different editor

Cells (editors after Schoots, 2023b, p. 71): Jabavu edits Isigidimi (Jan 1881–Oct 1884); Jabavu edits Imvo (Nov 1884–1888); Gqoba edits Isigidimi (Nov 1884–1888); Makiwane edits Isigidimi (1876–80; Stewart until 1876). Ratios of rates per million tokens; 95% CIs from 4000 issue resamples; p: two-sided, against 2000 matched sets of random words (src/15_editor_control.py). Every contrast is also a contrast in time; *rule* has only 9 uses in Jabavu's Isigidimi, so its large ratios are not interpreted.

Concept	Jabavu: Imvo / Isigidimi			Isigidimi: Gqoba / Jabavu			Isigidimi: 1881–84 / 1876–80		
	ratio	95% CI	p	ratio	95% CI	p	ratio	95% CI	p
parliament	1.69	1.23–2.54	0.088	0.50	0.33–0.79	0.043	49.86	14.41–405.79	<0.001
vote	3.07	1.98–5.23	0.017	2.36	1.07–4.61	0.034	11.63	4.32–98.99	0.004
government	1.21	0.99–1.49	0.387	0.59	0.44–0.77	0.043	0.78	0.61–1.01	0.289
rule	19.47	9.46–77.40	0.001	8.28	3.57–31.56	0.009	0.38	0.09–1.32	0.228
law	1.36	1.08–1.72	0.235	0.80	0.53–1.17	0.376	1.31	0.98–1.76	0.314
rights	0.93	0.76–1.16	0.808	0.82	0.58–1.13	0.480	2.91	2.05–4.12	0.006
country	1.08	0.92–1.29	0.773	1.13	0.90–1.41	0.596	0.74	0.59–0.92	0.243
land	1.10	0.91–1.34	0.689	0.95	0.70–1.28	0.853	0.94	0.76–1.16	0.815
freedom	0.68	0.51–0.92	0.443	0.95	0.64–1.38	0.899	1.52	1.01–2.42	0.408
humanity	0.97	0.59–1.90	0.959	1.88	0.98–3.88	0.227	0.97	0.45–1.96	0.967
religion	0.37	0.27–0.50	0.025	1.38	1.03–1.83	0.168	0.35	0.26–0.47	<0.001

S8. Hyper-parameter search for the embeddings

fastText skip-gram, tuned on text outside the test window (Isigidimi 1876–83, Imvo 1889–94) so that tuning never sees the compared years. Criterion: seed stability, the mean top-10 nearest-neighbour overlap between two runs that differ only in random seed (Antoniak & Mimno, 2018); unstable embeddings would turn noise into apparent shifts. 34 settings over dimension (50, 100, 200, 300, 400), window (1, 2, 3, 5, 8) and epochs (10, 20, 30), searched in rounds that extended the grid past each optimum. Chosen: dimension 400, window 1, 20 epochs. Stability is highest at the narrowest window, which favours grammar over topic; the paper therefore also reports windows 2 and 5 (Table S2). Best setting per window:

Window	Dimension	Epochs	Isigidimi 1876–83	Imvo 1889–94	Mean stability
1	400	20	0.844	0.814	0.829
2	300	20	0.823	0.798	0.810
3	300	20	0.810	0.799	0.805
5	400	20	0.798	0.796	0.797
8	200	10	0.758	0.755	0.757

At the chosen setting, word2vec reaches a stability of 0.64 against fastText's 0.83, so fastText is used throughout.

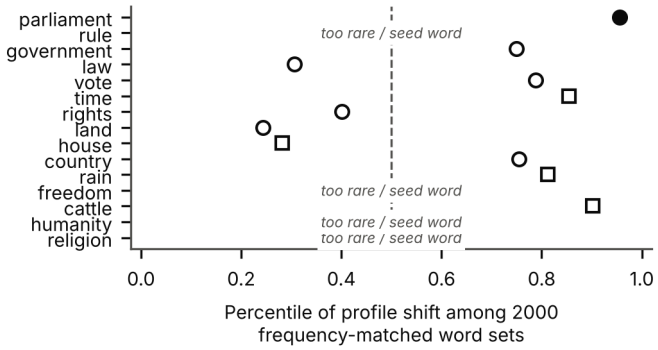


Figure S1: Profile-shift percentile of each concept among matched sets (markers as in Fig. 1, left); 'too rare' = too few occurrences, 'seed word' = anchor source, untestable.

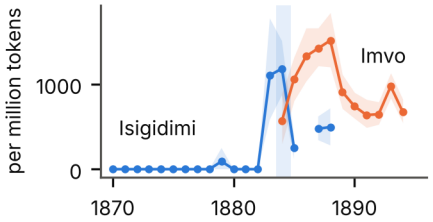


Figure S2: Parliamentary vocabulary per million tokens; it enters Isigidimi in 1883 (light blue: twice-monthly issues, Schoots et al., 2023a, p. 59).